

SoulScope Cross-Constellation Interaction Registry

v0.1

Status: Canonical scientific implementation companion — architecture final, interaction calibration pending

Governing authority: SoulScope Canon v1.3

Upstream: Constellation Dimension Registry v0.1; Inference Rule Registry v0.1; Constellation State Registry v0.1

Downstream: Pattern Registry; Narrative Registry

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NORTH STAR

Interactions describe relationships among independently resolved patterns. They do not create evidence, repair missing information, or imply causation.

1. Purpose

The Interaction Registry defines how independently completed SoulScope Constellations may be related after within-Constellation reasoning is finished.

The canonical sequence is:

Dimension Inference → Constellation Geometry → State / Blend / Unresolved → Interaction Engine → Pattern

The Interaction Engine may describe how two or more resolved functional structures coexist, align, oppose, modify, or change through time.

It SHALL NOT:

- rescore acoustic measurements;
 - modify Evidence;
 - change Dimension values;
 - replace an unresolved Dimension;
 - change a State decision;
 - infer causation;
 - create psychological facts not present upstream.
-

2. Interaction graph

The four Constellations form the nodes of a typed interaction graph:

$$G = (V, E)$$

where:

$$V = \{COG, REG, CAP, EXP\}$$

and each edge or hyperedge $e \in E$ represents a validated or research-authorized interaction.

Possible pairwise relationships:

$$COG \leftrightarrow REG$$

$$COG \leftrightarrow CAP$$

$$COG \leftrightarrow EXP$$

$$REG \leftrightarrow CAP$$

$$REG \leftrightarrow EXP$$

$$CAP \leftrightarrow EXP.$$

Higher-order interactions may include three or four Constellations.

3. Interaction input

The Interaction Engine receives the completed upstream objects:

```
InteractionInput

scan_id

constellations:
  COG:
    dimensions[]
    geometry
    state_or_blend
    qualifiers[]
    confidence
    contradiction

  REG:
```

```
dimensions[]
geometry
state_or_blend
qualifiers[]
confidence
contradiction
```

CAP:

```
dimensions[]
geometry
state_or_blend
qualifiers[]
confidence
contradiction
```

EXP:

```
dimensions[]
geometry
state_or_blend
qualifiers[]
confidence
contradiction
```

```
prompt_contexts[]
temporal_information[]
reference_information[]
```

```
registry_versions[]
```

The Engine SHALL reason primarily from **Dimension geometry and provenance**.

State labels provide context but are not sufficient by themselves to establish an interaction.

4. Why State names are insufficient

The relationship:

```
REG = STABLE_PATTERN
CAP = MAINTAINED_OUTPUT
```

does not reveal whether the scan contains:

```
Activation = high  
Effort Cost = low
```

or:

```
Activation = high  
Effort Cost = high.
```

Those configurations may support very different interaction interpretations.

Therefore:

```
Dimension geometry > State name alone
```

for Interaction reasoning.

5. Interaction object

Every candidate Interaction SHALL contain:

```
interaction_id  
interaction_version  
  
verb  
  
scientific_class  
scientific_status  
  
arity  
  
participant_constellations[]  
participant_dimensions[]  
  
source_role  
target_role  
  
interaction_axis  
  
prompt_scope  
temporal_scope  
reference_scope
```

```
required_dimension_conditions[]
optional_dimension_conditions[]

required_evidence_conditions[]

support
contradiction
coverage
confidence

candidate_fit

alternative_interactions[]

causal_status

publication_status
resolution_reason

rule_version
model_version
registry_versions[]
```

6. Causality boundary

No canonical Interaction verb establishes causal mechanism.

Therefore:

A Buffers B

means:

the observed configuration is consistent with a functional relationship in which the pattern represented by *A* accompanies lower observable cost or volatility associated with *B*.

It does **not** mean:

A biologically caused *B* to decrease.

Likewise:

A Destabilizes B

does not automatically establish causal destabilization.

HARD RULE

Interaction direction is functional direction within the registered model, not causal proof.

7. Interaction scientific classes

I1 – DESCRIPTIVE_RELATIONAL

Describes a directly observable relationship among already-resolved functional parameters.

Examples:

- Reinforces;
- Redirects;
- Shifts.

These still require validation but involve less additional semantic interpretation.

I2 – FUNCTIONAL_INTERPRETIVE

Adds a higher-order interpretation about cost, compensation, constraint, or regulation.

Examples:

- Buffers;
- Amplifies;
- Masks;
- Compensates;
- Constrains;
- Protects;
- Destabilizes.

These SHALL remain research-only until specifically validated.

I3 – HIGHER_ORDER_SYNTHESIS

Requires multi-Constellation or ambiguity-sensitive reasoning.

Examples:

- Integrates;
- Reveals.

These require explicit validation and strict traceability.

8. Scientific activation rule

Every interaction template begins as:

ENGINEERING_PRIOR

or:

RESEARCH_ONLY

until SoulScope data establish:

1. measurement reliability;
2. Dimension reliability;
3. recurring interaction structure;
4. construct validity where applicable;
5. calibration;
6. acceptable subgroup/generalization behavior.

No Interaction is scientifically validated merely because its semantic description appears reasonable.

9. Interaction axes

Interactions SHALL reference a registered functional axis rather than free-form semantic interpretation.

Initial axes:

AX_CONTINUITY
AX_ORGANIZATION
AX_MOBILIZATION
AX_STABILITY
AX_RECONFIGURATION
AX_MAINTENANCE

AX_COST
AX_MODULATION
AX_CONSTRAINT
AX_CONTEXTUAL_CHANGE
AX_COHERENCE

A future axis requires registry approval.

10. Interaction evidence principle

An Interaction requires independent upstream evidence for every participating Dimension.

The following is prohibited:

same acoustic feature $\rightarrow$ $Dimension_A \rightarrow Dimension_B \rightarrow Interaction$

when the apparent relationship is primarily caused by shared measurement redundancy.

Interaction confidence SHALL account for upstream Evidence independence.

11. Shared-source penalty

If two participating Dimensions depend heavily on the same Evidence families or source measurements, Interaction confidence shall be reduced.

Let:

$$Overlap(A, B) = \frac{|E_A \cap E_B|}{|E_A \cup E_B|}.$$

The exact overlap metric is calibration-dependent.

High source overlap does not invalidate an Interaction automatically, but it reduces claims of independent convergence.

12. Interaction fit

For candidate interaction I , the Engine evaluates a registered multidimensional relation region:

$$R_I.$$

Given participant Dimension posteriors $p_1, \dots, p_n$:

$$Fit_I = \int \cdots \int \left[\prod_{j=1}^n p_j(x_j) \right] \phi_I(x_1, \dots, x_n) dx_1 \cdots dx_n$$

where:

$$\phi_I \in [0, 1]$$

is the calibrated membership function for the interaction.

Until calibrated:

`candidate_fit` means compatibility with the interaction rule.

It SHALL NOT automatically be called probability.

13. Interaction confidence

Interaction confidence is distinct from fit.

Conceptually:

$$C_I = f(Fit_I, C_D, Coverage_I, Independence_I, Contradiction_I, TemporalSupport_I, Validation_I).$$

Where:

- C_D = participant Dimension confidence;
 - $Coverage_I$ = required Dimension/Evidence coverage;
 - $Independence_I$ = degree of independent upstream support;
 - $Contradiction_I$ = conflicting evidence;
 - $TemporalSupport_I$ = support for timing/direction where required;
 - $Validation_I$ = calibration status.
-

14. Interaction eligibility

A candidate Interaction is eligible only when:

1. all required participant Dimensions are resolved;
2. all required Dimensions are publication-eligible;
3. required prompt contexts exist;
4. required temporal information exists;

5. relevant source Evidence is sufficiently independent;
 6. no hard confound blocks comparison;
 7. the interaction template is active;
 8. its validated domain includes the active protocol.
-

15. Upstream abstention cannot be overridden

If:

REG-P4 Recovery = UNRESOLVED

no Interaction may infer Recovery indirectly.

Likewise:

CAP-P2 Reserve = UNRESOLVED

and:

EXP-P4 Relational Availability = UNRESOLVED

remain unavailable under the current protocol.

HARD RULE

Cross-Constellation reasoning cannot rescue a construct the protocol did not observe.

16. The canonical interaction vocabulary

The canonical verbs are:

1. Reinforces
2. Buffers
3. Amplifies
4. Masks
5. Compensates
6. Constrains
7. Protects
8. Redirects

- 9. Destabilizes
- 10. Integrates
- 11. Reveals
- 12. Shifts

No implementation may invent additional user-semantic interaction verbs without registry governance.

17. INT-REINFORCES – Reinforces

Class: I1

Typical arity: 2

Directionality: usually symmetric

Canonical meaning:

Two independently resolved structures support the same bounded functional interpretation.

Conceptually:

$$A \uparrow + B \uparrow \Rightarrow \text{shared functional tendency strengthened}$$

but only when the dimensions share a registered Interaction axis.

Example candidate hypotheses

COG × REG

COG Organization ↑
REG Stability ↑

may reinforce:

AX_CONTINUITY / AX_ORGANIZATION

REG × CAP

REG Activation ↑
CAP Mobilization ↑

may reinforce:

AX_MOBILIZATION

EXP × REG

EXP Range ↑
REG Flexibility ↑

may reinforce:

AX_RECONFIGURATION

after both constructs are validated.

Exclusion

Similar numeric directions do not automatically constitute reinforcement.

They must share a registered functional axis.

18. INT-BUFFERS – Buffers

Class: I2

Typical arity: 2

Directionality: asymmetric

Canonical structure:

buffer_source
target_demand_or_volatility

Meaning

One independently supported pattern coexists with lower observable cost, disruption, or volatility than might otherwise be expected under the registered relation.

This is a functional comparative interpretation.

It is not proof of a biological buffering mechanism.

Candidate hypotheses

REG → COG

COG Processing Demand ↑
REG Stability ↑
COG Focus Continuity maintained

Candidate interpretation:

REG Stability
BUFFERS
observable disruption associated with Processing Demand

CAP → COG

COG Processing Demand ↑
CAP Effort Cost lower
CAP Sustainability maintained

Candidate:

CAP pattern BUFFERS demand-associated cost

Required logic

A Buffer interaction requires evidence of:

1. a genuine target demand/volatility signal; and
2. preserved function or lower cost in another independent domain.

No target demand:

⇒ no Buffer interaction.

19. INT-AMPLIFIES – Amplifies

Class: I2

Typical arity: 2

Directionality: asymmetric or mutually reinforcing depending template

Meaning

Two independently resolved patterns coincide in a way associated with greater observable intensity, cost, or disruption.

It does not mean one pattern caused the other.

Candidate hypotheses

COG × CAP

COG Processing Demand ↑
CAP Effort Cost ↑

Candidate:

AMPLIFIES
AX_COST

REG × CAP

REG Activation ↑
CAP Effort Cost ↑

may support an Amplifies candidate if the combination is associated with greater observable output cost.

COG × REG

COG Processing Demand ↑
REG Stability ↓

may support:

demand-associated pattern
AMPLIFIES
observable volatility

after validation.

Required distinction

two high values / Amplifies. =

The joint configuration must add explanatory value beyond its separate components.

20. INT-MASKS – Masks

Class: I2

Typical arity: 2

Directionality: asymmetric

Canonical meaning

Comparatively maintained or adaptive-looking observable function in one Constellation coexists with independently supported cost, constraint, or disruption in another.

HARD SEMANTIC BOUNDARY

Masks does **not** mean:

- concealment;
- dishonesty;
- suppression of truth;
- hiding emotion;
- conscious compensation;
- hidden psychological state.

It describes **cross-domain mismatch**, not deception.

Preferred internal name

CROSS_DOMAIN_MAINTENANCE_COST_MISMATCH

The user-facing verb may remain **Masks** only if comprehension testing demonstrates that users do not interpret it as concealment.

Candidate hypotheses

COG × CAP

COG Organization ↑
COG Focus Continuity ↑

CAP Effort Cost ↑

Candidate interpretation:

maintained speech-production organization coexisted with elevated observable cost.

EXP × CAP

EXP Range broad
CAP Effort Cost ↑

may support a mismatch interaction where expressive modulation remains broad despite independently evidenced output cost.

Important

No “hidden” property is inferred.

The second Constellation's cost is explicitly measured by its own evidence chain.

21. INT-COMPENSATES – Compensates

Class: I2

Typical arity: 2

Directionality: asymmetric

Canonical roles:

strength_domain
limitation_domain

Meaning

A comparatively strong functional pattern coexists with preserved output despite an independently supported limitation elsewhere.

It does not establish a conscious strategy or causal compensation.

Candidate hypothesis

COG × CAP

CAP Sustainability ↓
COG Focus Continuity maintained
COG Organization maintained

may support the hypothesis that preserved Cognitive Form is compatible with a compensatory relation despite reduced short-interval maintenance elsewhere.

REG × COG

COG Organization lower
REG Stability higher

may become a candidate only if validation shows that stable regulation reliably accompanies preserved function despite organizational limitation.

Required condition

There must be:

limitation + preserved function + independent strength

not merely one high and one low Dimension.

22. INT-CONSTRAINS – Constrains

Class: I2

Typical arity: 2

Directionality: asymmetric

Meaning

A limitation in one independently measured functional domain is associated with reduced expression or movement in another domain under compatible context changes.

Directionality requires temporal/contextual evidence.

Candidate hypothesis

CAP → EXP

CAP Sustainability ↓
EXP Range ↓

alone is insufficient.

More compelling research evidence would be:

$\Delta CAP_{\text{maintenance}} \downarrow$

co-occurring systematically with:

$\Delta EXP_{\text{deployment}} \downarrow$

across compatible prompt transitions.

Hard boundary

Without temporal/contextual evidence:

CONSTRAINS

SHALL NOT be inferred from static co-occurrence alone.

Static association may be retained as research evidence without directional labeling.

23. INT-PROTECTS – Protects

Class: I2

Typical arity: 2

Directionality: asymmetric

Meaning

A stable or bounded functional pattern coexists with lower observable strain, disruption, or cost under a context in which those outcomes could otherwise have appeared.

This does not establish biological protection.

Candidate hypotheses

REG → CAP

REG Stability ↑
CAP Effort Cost lower
CAP Sustainability maintained

under elevated Activation or Mobilization may support a Protects candidate.

COG → CAP

COG Organization maintained
CAP Effort Cost lower

under higher Processing Demand may support a research candidate.

Required condition

Protects requires a meaningful **challenge/demand context**.

Stable + low cost in an ordinary condition is insufficient.

24. INT-REDIRECTS – Redirects

Class: I1

Typical arity: 2

Directionality: asymmetric

Temporal requirement: strong

Meaning

A context transition is accompanied by reduced expression in one functional channel and increased expression in another.

“Energy” is metaphorical product language only.

The scientific object is:

opposing coordinated cross-domain change.

Candidate hypothesis

REG × EXP

Across:

$$P1 \rightarrow P2$$

suppose:

REG Activation ↑
EXP Range ↓

with coherent independent evidence.

This may support:

REDIRECTS

as a research hypothesis describing increased mobilization alongside reduced outward modulation.

The system SHALL NOT say:

emotional energy was redirected inward.

That would exceed the evidence.

25. INT-DESTABILIZES – Destabilizes

Class: I2

Typical arity: 2

Directionality: asymmetric

Temporal requirement: required for directional publication

Meaning

Change in one functional domain is associated with a subsequent or concurrent reduction in coherence/stability in another under the registered temporal rule.

Candidate hypothesis

Across prompt transition:

COG Processing Demand ↑
REG Stability ↓
REG within-prompt volatility ↑

may support a research hypothesis of a destabilizing relation.

Hard rule

Static combination:

Processing Demand high
Stability low

alone does not prove:

Processing Demand DESTABILIZES Stability.

Without temporal support, the system may only report co-occurrence.

26. INT-INTEGRATES – Integrates

Class: I3

Preferred arity: 3 or 4

Directionality: symmetric higher-order relation

Meaning

Multiple independently resolved Constellations align coherently, with low contradiction and manageable observable cost.

This is not a claim of psychological integration or health.

Candidate structure

For example:

COG Organization maintained
REG Stability maintained
REG Flexibility adequately responsive
CAP Sustainability maintained
CAP Effort Cost not elevated
EXP modulation appropriately deployed

may support an **INTEGRATES** candidate after calibration.

Required components

At minimum:

3 Constellations

plus:

- low cross-domain contradiction;
- sufficient Evidence independence;
- no major high-cost contradiction;
- coherent contextual behavior.

Prohibited interpretation

Integrates SHALL NOT mean:

healthy
whole
healed
emotionally integrated
psychologically optimal
spiritually aligned

27. **INT-REVEALS** – Reveals

Class: I3

Typical arity: 2

Directionality: asymmetric

Canonical roles:

```
clarifying_constellation  
ambiguous_constellation
```

Meaning

A second independently resolved Constellation provides information that narrows the appropriate interpretation of an otherwise ambiguous or boundary-positioned first Constellation.

HARD RULE

Reveals does not change the first Constellation's:

- Dimension values;
- State;
- Blend;
- unresolved status.

It clarifies **interpretation**, not upstream truth.

Example

Suppose:

```
COG = boundary blend  
between more continuous and more fragmented organization
```

while:

```
CAP Effort Cost ↑
```

is strongly resolved.

The Interaction Engine may record:

```
CAP REVEALS cost context relevant to interpreting COG ambiguity
```

The Narrative may say:

The mixed Cognitive Form pattern appeared alongside higher observable effort cost.

It SHALL NOT retroactively select one COG State.

28. INT-SHIFTS – Shifts

Class: I1

Arity: 2+ over time

Directionality: temporal

Meaning

The relationship among Constellations changes from one interaction regime to another over time.

Possible scopes:

```
WITHIN_PROMPT  
P1_TO_P2  
P2_TO_P3  
P1_TO_P3  
LONGITUDINAL
```

Representation

$$I(t_1) \rightarrow I(t_2)$$

Example:

```
P1:  
  REG × CAP = low-cost stable relation  
  
P2:  
  REG × CAP = higher-activation / higher-cost relation  
  
P3:  
  REG × CAP = maintained output with lower cost
```

The system may record:

```
interaction = SHIFTS
```

without asserting emotional improvement or recovery.

29. Interaction directionality classes

Every interaction uses one of:

```
SYMMETRIC  
FUNCTIONALLY_DIRECTED  
TEMPORALLY_DIRECTED  
HYPEREDGE
```

Symmetric

Example:

```
REINFORCES
```

when neither participant is privileged.

Functionally directed

Example:

```
REG Stability BUFFERS COG disruption
```

The direction belongs to the operational model.

It is not causal proof.

Temporally directed

Requires time ordering.

Example:

```
A change precedes or accompanies B under registered timing rule.
```

Hyperedge

Three or more participants.

Example:

INTEGRATES

30. Pairwise candidate-template registry

v0.1 establishes the following research template domains.

These are **candidate spaces**, not automatically active interactions.

COG × REG

Candidate dimensions:

COG Organization
COG Focus Continuity
COG Processing Demand

REG Activation
REG Stability
REG Flexibility

Candidate verbs:

REINFORCES
BUFFERS
AMPLIFIES
COMPENSATES
DESTABILIZES
REVEALS
SHIFTS

31. COG × CAP

Candidate dimensions:

COG Organization
COG Focus Continuity
COG Processing Demand

CAP Mobilization
CAP Effort Cost
CAP Sustainability

Candidate verbs:

REINFORCES
BUFFERS
AMPLIFIES
MASKS
COMPENSATES
PROTECTS
REVEALS
SHIFTS

This pairing is expected to be especially important for distinguishing:

functional output

from:

cost of maintaining that output.

32. COG × EXP

Candidate dimensions:

COG Organization
COG Exploration
COG Focus Continuity
COG Processing Demand

EXP Range
EXP Openness
EXP Restraint

Candidate verbs:

REINFORCES
CONSTRAINS
REDIRECTS

REVEALS
SHIFTS

Because Exploration, Openness, and Restraint are D2 constructs, many of these mappings remain research-only until those constructs are validated.

33. REG × CAP

Candidate dimensions:

REG Activation
REG Stability
REG Flexibility

CAP Mobilization
CAP Effort Cost
CAP Sustainability

Candidate verbs:

REINFORCES
BUFFERS
AMPLIFIES
PROTECTS
DESTABILIZES
INTEGRATES
SHIFTS

This pairing is expected to be particularly useful for distinguishing:

mobilization

from:

cost and maintenance.

34. REG × EXP

Candidate dimensions:

REG Activation
REG Stability
REG Flexibility

EXP Range
EXP Openness
EXP Restraint

Candidate verbs:

REINFORCES
CONSTRAINS
REDIRECTS
DESTABILIZES
REVEALS
SHIFTS

No relational/attachment inference may be introduced through this pairing.

35. CAP × EXP

Candidate dimensions:

CAP Mobilization
CAP Effort Cost
CAP Sustainability

EXP Range
EXP Openness
EXP Restraint

Candidate verbs:

REINFORCES
MASKS
COMPENSATES
CONSTRAINS
PROTECTS
REDIRECTS

REVEALS
SHIFTS

36. Current unavailable interaction inputs

The current scan SHALL NOT use:

REG-P4 Recovery
CAP-P2 Reserve
EXP-P4 Relational Availability

inside any production interaction.

Therefore candidate templates depending on them remain inactive.

37. Prompt-context interaction logic

Interaction evidence may exist separately in each prompt.

Represent:

$$I_{P1}, I_{P2}, I_{P3}.$$

This lets the system distinguish:

a relationship that appears globally

from:

a relationship that appears only when the context changes.

A whole-scan interaction summary SHALL preserve its prompt composition.

38. Prompt-specific interaction record

PromptInteraction

prompt_id

```
participant_dimensions[]  
  
interaction_candidate  
  
fit  
confidence  
  
support  
contradiction  
  
evidence_trace[]
```

39. Cross-prompt interaction record

```
ContextTransitionInteraction  
  
source_prompt  
target_prompt  
  
participant_dimensions[]  
  
delta_a  
delta_b  
  
temporal_relation  
  
candidate_verb  
  
fit  
confidence  
  
alternative_candidates[]
```

40. Static versus dynamic interaction

Every interaction SHALL declare:

```
STATIC_COEXISTENCE  
CONTEXTUAL_RELATION
```

TEMPORAL_TRANSITION
LONGITUDINAL_RELATION

Certain verbs require a minimum type.

For example:

REDIRECTS
-> CONTEXTUAL_RELATION or TEMPORAL_TRANSITION

SHIFTS
-> TEMPORAL_TRANSITION or LONGITUDINAL_RELATION

CONSTRAINS
-> preferably CONTEXTUAL_RELATION or stronger

DESTABILIZES
-> TEMPORAL_TRANSITION required for directional language

41. Interaction contradiction

An interaction candidate may be contradicted even when its participant Dimensions are individually well resolved.

Example:

Candidate:
REG Stability BUFFERS COG Processing Demand

but:

COG Focus Continuity ↓ strongly
CAP Effort Cost ↑ strongly

may contradict the Buffer interpretation.

The Interaction Ledger must preserve these cross-domain contradictions.

42. Alternative Interaction candidates

For each evaluated participant set, the Engine SHALL store plausible alternatives.

Example:

```
primary:
  AMPLIFIES

alternative:
  REINFORCES

reason:
  both Dimensions moved in the same direction,
  but the joint configuration was more strongly
  associated with increased observable cost.
```

Until calibration, this remains research logic.

43. Interaction competition

Multiple verbs may describe different axes involving the same two Constellations.

Example:

```
REG × CAP

REINFORCES:
  mobilization axis

AMPLIFIES:
  cost axis
```

Both may coexist if they describe distinct registered relations.

However, for the **same participant dimensions + same axis + same temporal scope**, the Engine SHOULD normally publish one primary interaction or abstain if competing candidates are insufficiently separated.

44. User-facing interaction limit

The backend may retain many valid interactions.

The user-facing result SHOULD normally surface no more than:

2

primary Interaction observations in the main result.

Additional interactions may appear through progressive disclosure.

This protects interpretability without reducing backend richness.

45. Interaction significance is not dramatic language

The Pattern/Narrative system SHALL rank interactions by:

- evidence strength;
- confidence;
- explanatory contribution;
- independence;
- user relevance.

It SHALL NOT rank them by how psychologically dramatic the verb sounds.

46. Interaction abstention reasons

Canonical reasons:

```
PARTICIPANT_DIMENSION_UNRESOLVED  
PARTICIPANT_NOT_PRODUCTION_ACTIVE  
INSUFFICIENT_EVIDENCE_INDEPENDENCE  
INSUFFICIENT_TEMPORAL_SUPPORT  
MISSING_REQUIRED_CONTEXT  
HIGH_CROSS_DOMAIN_CONTRADICTION  
INTERACTION_MODEL_NOT_VALIDATED  
INTERACTION_FIT_TOO_LOW  
COMPETING_INTERACTIONS_UNRESOLVED  
MODEL_OUT_OF_DISTRIBUTION
```

```
ACTIVE_HARD_CONFOUND  
PROTOCOL_UNOBSERVABLE
```

47. Interaction publication states

```
PUBLISHED  
PUBLISHED_WITH_CAUTION  
RESEARCH_ONLY  
UNRESOLVED  
SUPPRESSED  
INVALID
```

48. Interaction Decision Ledger

Every candidate evaluation creates:

```
InteractionDecisionRecord  
  
interaction_decision_id  
scan_id  
  
participant_constellations[]  
participant_dimensions[]  
  
verb_candidate  
  
interaction_axis  
temporal_scope  
  
hard_gates[]  
gate_results[]  
  
source_dimension_decision_ids[]  
  
source_state_decision_ids[]  
  
supporting_evidence[]  
contradicting_evidence[]  
  
shared_source_overlap
```

```
required_conditions_passed[]
required_conditions_failed[]

candidate_fit
confidence_components

alternative_interactions[]

selected_interaction

publication_status

primary_reason
secondary_reasons[]

rule_version
model_version
registry_versions[]

created_at
```

49. No silent shared-source inflation

The Decision Ledger SHALL identify when two Dimensions rely materially on overlapping acoustic evidence.

Example:

```
shared_evidence_families:
  PRO
  DYN

independence_penalty:
  applied
```

This prevents one acoustic phenomenon from appearing falsely as multiple independent confirmations.

50. Interaction evidence trace

For a published interaction:

Interaction $\leftarrow$ *Dimensions* $\leftarrow$ *Evidence* $\leftarrow$ *Measurements*.

A trace view must answer:

1. Which Constellations participated?
2. Which Dimensions mattered?
3. Which prompts/context produced the relation?
4. Which Evidence supported it?
5. Which Evidence contradicted it?
6. How independent were the sources?
7. What alternatives were evaluated?
8. Why did this verb win?
9. Was temporal direction actually observed?
10. Which rule/model versions produced the conclusion?

51. Example — Reinforces

Illustrative research example:

REG-P1 Activation:

higher

confidence: .82

CAP-P1 Mobilization:

higher

confidence: .79

Both have independently qualified evidence.

Candidate:

verb:

REINFORCES

axis:

AX_MOBILIZATION

Explanation:

Regulatory activation and observable output mobilization moved together in this scan.

Not:

You were highly emotionally activated.

52. Example — Buffers

Illustrative:

COG-P4 Processing Demand:
higher

REG-P2 Stability:
higher

COG-P3 Focus Continuity:
maintained

CAP-P3 Effort Cost:
not elevated

Research candidate:

REG Stability
BUFFERS
demand-associated disruption

The backend may test this candidate.

The user-facing result SHALL not imply physiological protection unless separately validated.

53. Example — Amplifies

Illustrative:

COG Processing Demand:
higher

CAP Effort Cost:
higher

Timing fragmentation:
increased

Candidate:

AMPLIFIES
AX_COST

Possible bounded narrative after validation:

The increased speech-production demand appeared alongside greater observable cost, making that pattern more pronounced.

54. Example — Masks

Illustrative:

COG Organization:
higher

COG Focus Continuity:
higher

CAP Effort Cost:
higher

Candidate internal result:

CROSS_DOMAIN_MAINTENANCE_COST_MISMATCH

Potential user translation:

Your responses remained organized even while the acoustic pattern showed greater observable cost.

This is preferable to:

You were masking strain.

55. Example — Compensates

Illustrative:

CAP Sustainability:
lower

COG Focus Continuity:
maintained

COG Organization:
maintained

Research candidate:

preserved speech-production organization coexisted with lower short-interval maintenance.

The verb **Compensates** becomes publishable only if longitudinal/experimental validation supports the higher-order interpretation.

56. Example — Redirects

Across:

$$P1 \rightarrow P2$$

suppose:

REG Activation:
rises

EXP Range:
narrows

EXP deployment:
decreases

Evidence:
independent and coherent

Candidate:

REDIRECTS

Bounded narrative:

As the context changed, mobilization increased while outward acoustic modulation became narrower.

Not:

Your emotional energy moved inward.

57. Example — Integrates

Illustrative future candidate:

COG:
organized
continuous

REG:
stable
context-responsive

CAP:
maintained
low-moderate cost

EXP:
context-responsive modulation

cross-domain contradiction:
low

Candidate:

INTEGRATES

This means:

several independently measured functional patterns aligned coherently.

It does not mean the person is psychologically integrated.

58. Example — Reveals

Suppose:

```
COG State:
  boundary blend

CAP:
  Effort Cost higher
  confidence high
```

Interaction:

```
CAP REVEALS context relevant to COG ambiguity
```

Narrative:

The mixed Cognitive Form pattern occurred alongside greater observable cost, which adds context to why the scan did not resolve into a single Cognitive Form state.

The COG blend remains unchanged.

59. Example — Shifts

Within one scan:

```
P1:
  REG × CAP
  stable / low-cost relation

P2:
  REG × CAP
  higher activation + higher cost

P3:
```

REG × CAP
maintained output + lower cost

Result:

SHIFTS

Narrative:

The relationship between mobilization and cost changed across the three speaking contexts.

No recovery inference is allowed.

60. Multi-interaction pattern

A completed scan may contain:

COG × CAP:
AMPLIFIES

REG × CAP:
BUFFERS

REG × EXP:
REDIRECTS

These are not contradictions merely because they use different verbs.

Each describes a different participant set or functional axis.

The Pattern Engine determines which relationships are most useful to summarize.

61. Interaction model strategy

v0.1 SHALL begin with transparent rule templates.

Preferred initial architecture:

registered Dimension regions + posterior overlap + temporal relation + independence + contradiction → candidat

Do not begin with an opaque model trained to produce interaction labels directly.

62. Interaction discovery research

As SoulScope accumulates validated data, candidate relationships may be discovered through:

- hierarchical models;
- multilevel regression;
- temporal models;
- graph models;
- latent interaction analysis;
- clustering of cross-Constellation trajectories.

Discovery models may propose candidates.

They do not activate them.

63. Interaction promotion process

Candidate → Replication → Reliability → Construct validation → Incremental value → Calibration → Comprehension

64. Incremental-value requirement

Before activating Interaction I , test whether it explains something materially useful beyond reporting its participant Dimensions separately.

If:

$$Narrative(D_A, D_B)$$

communicates the result just as accurately and understandably as:

$$Interaction_I(D_A, D_B),$$

then the interaction verb may not add sufficient value to justify production activation.

65. User comprehension requirement

Interaction verbs can invite overinterpretation.

Therefore comprehension studies SHALL specifically test whether users interpret:

```
Buffers
Masks
Compensates
Protects
Constrains
Destabilizes
```

as causal or psychological certainties.

If misunderstanding persists:

- wording must change;
- interaction may remain internal;
- or the verb may be replaced in user-facing narrative while retaining the canonical internal relation ID.

66. Internal name versus user language

Every Interaction SHALL maintain:

```
canonical_verb
internal_operational_name
user_language_key
```

Example:

```
canonical_verb:
  Masks

internal_operational_name:
  CROSS_DOMAIN_MAINTENANCE_COST_MISMATCH

user_language_key:
  maintained_output_with_cost
```

This permits scientifically safer user wording without changing backend logic.

67. Interaction desirability

No Interaction has intrinsic positive or negative value.

For example:

BUFFERS

is not automatically healthy.

AMPLIFIES

is not automatically bad.

INTEGRATES

is not automatically ideal.

Meaning remains scan-bound and context-bound.

68. Interaction and personal baseline

With longitudinal history, the Engine may evaluate:

$$I_t$$

relative to the person's prior interaction distribution.

Possible longitudinal observations:

NEW_INTERACTION
PERSISTENT_INTERACTION
REDUCED_INTERACTION_SUPPORT
INCREASED_INTERACTION_SUPPORT
INTERACTION_REGIME_CHANGE
INSUFFICIENT_COMPARABILITY

These remain descriptive.

69. Longitudinal Shifts

Longitudinal Shifts requires compatible:

- protocol;
- feature registry;
- Dimension model;
- Interaction model;
- baseline/reference rules.

If compatibility fails:

```
SHIFTS = UNRESOLVED
```

rather than comparing unlike versions.

70. Pattern Engine handoff

The Interaction Engine produces:

```
InteractionResultSet

published_interactions[]
caution_interactions[]
research_interactions[]

unresolved_candidates[]

cross_domain_contradictions[]

temporal_regime_changes[]

interaction_decision_records[]

registry_version
```

The Pattern Engine may consume only publication-authorized interactions.

71. Narrative Engine handoff

Narrative receives:

```
interaction_id
participant_dimensions
bounded_meaning
confidence
temporal_scope
support_summary
contradiction_summary
user_language_key
prohibited_claims[]
```

It SHALL NOT create a stronger causal or emotional explanation than this object permits.

72. Acceptance tests

INT-UPSTREAM-001

Given one required participant Dimension unresolved:

Expected:

```
interaction = UNRESOLVED
```

No inferred replacement.

INT-D3-001

Given an Interaction requiring Recovery under current protocol:

Expected:

```
INELIGIBLE
```

INT-CAUSAL-001

Given static co-occurrence only:

Expected:

DESTABILIZES

cannot be published directionally.

INT-REDIRECT-001

Given opposing cross-context changes in two independently supported domains:

Expected:

REDIRECTS may become an eligible research candidate.

INT-MASK-001

Given maintained COG function plus elevated CAP Effort Cost:

Expected internal candidate:

CROSS_DOMAIN_MAINTENANCE_COST_MISMATCH

No deception/concealment inference.

INT-INDEP-001

Given two Dimensions primarily derived from the same acoustic Evidence:

Expected:

Interaction independence decreases.

No artificial multi-domain confirmation.

INT-REVEAL-001

Given a boundary-blended COG State and strongly resolved CAP context:

Expected:

REVEALS may annotate interpretation.

COG State remains unchanged.

INT-SHIFT-001

Given interaction regime differs between P1 and P2:

Expected:

SHIFTS may be evaluated.

No Recovery inference.

INT-MULTI-001

Given several valid interactions on different axes:

Expected:

All may remain in backend result.

Pattern/Narrative prioritization occurs downstream.

INT-LLM-001

Given narrative model attempts to invent an unregistered interaction:

Expected:

architecture violation.

73. Hard invariants

INT - INV - 01

Interactions occur only after independent Constellation reasoning.

INT - INV - 02

Interactions do not create or modify Dimensions.

INT - INV - 03

Interactions do not resolve D3 constructs.

INT - INV - 04

Interaction direction is not causal proof.

INT - INV - 05

State labels alone are insufficient for interaction reasoning.

INT - INV - 06

Source-Evidence overlap must be measured.

INT - INV - 07

Static co-occurrence cannot support temporally directional verbs without appropriate temporal evidence.

INT - INV - 08

Masks does not imply concealment.

INT - INV - 09

Integrates does not imply psychological health or wholeness.

INT - INV - 10

Reveals cannot rewrite upstream ambiguity.

INT - INV - 11

Shifts describes change of relation, not improvement or recovery.

All interaction thresholds and region models remain provisional until calibrated.

74. What v0.1 freezes

The following are canonical:

- typed cross-Constellation graph architecture;
 - 12 canonical interaction verbs;
 - interaction scientific classes;
 - functional-axis requirement;
 - Dimension-first reasoning;
 - State labels as contextual rather than sufficient input;
 - explicit Evidence-independence accounting;
 - causal boundary;
 - static versus temporal interaction distinction;
 - pairwise and higher-order interaction support;
 - prompt-specific interaction storage;
 - alternative-candidate storage;
 - abstention;
 - Decision Ledger traceability;
 - D3 prohibition;
 - Pattern/Narrative separation;
 - longitudinal interaction compatibility requirements.
-

75. What v0.1 intentionally does not freeze

The following require SoulScope data:

- exact interaction-region definitions;
 - posterior thresholds;
 - minimum interaction fit;
 - minimum confidence;
 - source-overlap penalties;
 - specific pairwise weightings;
 - which candidate templates recur reliably;
 - which I2/I3 interactions earn production activation;
 - final user-facing wording;
 - longitudinal significance thresholds.
-

76. Recommended initial implementation status

Architectural production support

Implement the engine, schemas, Decision Ledger, abstention, and trace behavior now.

Research candidate interaction set

Begin collecting evidence for:

REINFORCES
BUFFERS
AMPLIFIES
MASKS
COMPENSATES
CONSTRAINS
PROTECTS
REDIRECTS
DESTABILIZES
INTEGRATES
REVEALS
SHIFTS

but mark unvalidated mappings:

RESEARCH_ONLY

until earned.

This allows SoulScope to learn the interaction system without prematurely presenting hypotheses as established findings.

77. Recommended research priority

The first interaction families to test should be the ones most directly supported by the current three-prompt architecture.

Priority 1

REINFORCES
REDIRECTS
SHIFTS

because they can be framed primarily as observable relational/contextual structure.

Priority 2

BUFFERS
AMPLIFIES
MASKS

because the distinction between function, mobilization, and measurable cost is central to SoulScope.

Priority 3

COMPENSATES
CONSTRAINS
PROTECTS
DESTABILIZES

because their semantics require stronger functional and temporal validation.

Priority 4

INTEGRATES
REVEALS

because they are higher-order synthesis constructs and should be activated only after the underlying interaction system is stable.

78. Why this layer matters

Without an Interaction Engine, SoulScope would produce four parallel summaries.

With a valid Interaction Engine, it can distinguish configurations such as:

high demand + maintained output + low cost

from:

high demand + maintained output + high cost

and:

high activation + broad modulation

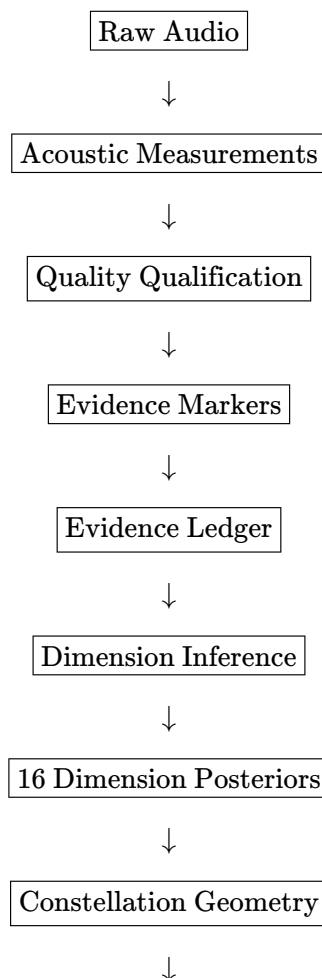
from:

high activation + compressed modulation.

That is where much of SoulScope's semantic richness can emerge without inventing hundreds of named states.

79. Complete semantic architecture

The semantic backend is now:



State / Blend / Unresolved + Qualifiers



Cross-Constellation Interactions



Pattern Engine



Narrative

At every level:

Decision Ledger + Provenance

remains reconstructable.

80. Final Interaction law

SoulScope may describe how independently measured patterns relate.

It may not turn relationship into causation, coexistence into hidden truth, or semantic elegance into evidence.

Independent evidence + resolved geometry + registered relationship + uncertainty → bounded interaction

When that chain is incomplete:

No interaction is the correct interaction.